

Chapter 1

Understanding a business

1.1 Goals, revenue and profit

A business creates value by offering goods or services that customers want. A for-profit business also aims to earn a return for its owners and remain financially viable. Other organisations can pursue a public, social or environmental mission instead of distributing profit to owners.

Revenue is earned from selling goods or services in the ordinary course of business. *Profit* is the result after deducting the relevant expenses for a period. A business can therefore record revenue and still make a loss. Profit also differs from cash: a sale can generate revenue before the customer pays.

In a simplified model,

$$\text{Profit or loss} = \text{Revenue} - \text{Expenses}.$$

If expenses exceed revenue, the result is a loss. To interpret this calculation, revenue and expenses must refer to the same period.

1.2 Resources and stakeholders

Four useful categories of business resources are financial resources (money and access to funding), human resources (people and skills), technology (tools, systems and know-how), and physical resources (materials, equipment and facilities). They work together; none is necessarily the most important in every business.

Stakeholders are people or groups who affect the organisation or are affected by it. They include owners or shareholders, executives, managers, employees, customers, suppliers, lenders, public authorities and the community. Their interests may differ. For example, owners may seek a return, employees may value pay and job security, and customers may seek quality and a fair price. These are typical interests, not fixed rules; some roles can overlap.

1.3 Money flows

Cash can enter a business through customer payments, owner contributions or loans. It can leave through payments to suppliers, employees, tax authorities and lenders, or when the business invests in equipment. Owner contributions and loans increase cash but are not sales revenue. Buying equipment with cash reduces cash and normally creates an asset; it is not necessarily an expense of the same amount at that instant. For these reasons, cash movements alone do not determine the period's profit.

1.4 Types of organisations and changes over time

For-profit companies seek returns for their owners and may reinvest or distribute profits. Non-profit organisations use any surplus to support their mission. Cooperatives are owned by members and serve their shared interests. Social enterprises use business activity to pursue social or environmental goals. Government agencies provide public services rather than operating primarily to make a profit. These categories describe aims or ownership and can overlap; they are not all legal forms.

A new business may face uncertainty and spend heavily before becoming profitable. If it builds a customer base and a workable business model, some risks may decline and profit may improve. This is not automatic. Competition, technological change, debt or a saturated market can later increase risk or reduce profitability. There is no universal path by which every company becomes safer and more profitable with age.

1.5 Legal forms, country rules and size

A freelancer works independently for clients. Freelancing is a way of working, not one universal legal form. A sole proprietor or sole trader operates a business alone, generally without creating a separate company entity and with personal liability for business obligations. A partnership has two or more owners; their liability depends on the form of partnership and the law. A limited-liability company or corporation is a separate entity whose owners generally have limited liability, subject to exceptions. Such an entity may have one owner or many: limited liability does not mean ownership by a small group.

Legal forms and reporting duties vary by country. *Inc.* is used in the United States, *AG* in Germany and *S.p.A.* in Italy; these designations should not be treated as interchangeable. Registration, tax and employment rules also depend on where the business operates.

As a company grows, coordinating more employees, products, customers and markets can increase management, communication and control costs. At the same time, a larger scale can reduce the cost per unit through economies of scale. Total costs and costs per unit should therefore be considered separately.

Chapter 2

Finance and accounting

2.1 Finance and financial information

Finance concerns how a business obtains, uses and manages funds. It addresses questions such as how much cash is available, how much funding is needed, whether to borrow or raise capital, and which investments to make. Financial accounting reports performance and financial position; management or cost accounting provides information for internal decisions. Taxation, financial management and investment analysis are related areas. Finance and accounting use some of the same information, but finance is not another name for bookkeeping.

2.2 Financial statements and key terms

The balance sheet, also called the statement of financial position, reports assets, liabilities and equity at a specific date. The income statement, or statement of profit or loss, reports revenue, expenses and the resulting profit or loss over a period. The cash flow statement reports how cash and cash equivalents changed over a period through operating, investing and financing activities. These are useful starting points; a complete set of statements may contain additional reports and notes.

Revenue is earned from ordinary business activities and need not have been collected in cash yet. Profit is the result after the relevant expenses are deducted, not a cash balance. Cash is money on hand or available in a bank account. Debt is borrowing that must be repaid and is one type of liability; unpaid supplier invoices are liabilities too. Equity is the owners' residual interest in the business. An owner contribution increases equity without creating revenue.

2.3 The accounting equation

The relationship between the main elements of the balance sheet is

$$\text{Assets} = \text{Liabilities} + \text{Equity}.$$

Assets are economic resources controlled by the business. Liabilities are present obligations to others. Equity is what remains of the assets after deducting the liabilities. The equation remains balanced after each correctly recorded transaction. Revenue increases equity through profit, expenses reduce it, and owner contributions can increase it without being revenue.

For example, if an owner invests EUR 1,000 in cash, cash and equity each increase by EUR 1,000. If the business then buys a machine for EUR 500 in cash, the machine asset increases and the cash asset decreases by EUR 500: total assets do not change. If it buys the machine on credit instead, the machine asset and an amount payable to the supplier each increase by EUR 500. Paying the supplier later decreases both cash and the payable by EUR 500.

Consider a car purchased for EUR 30,000 with payment due in 60 days. On purchase, the car asset and a payable each increase by EUR 30,000. On payment, cash and the payable each decrease by EUR 30,000. If the car is paid for immediately, the car asset increases and cash decreases. The purchase itself is generally not an immediate EUR 30,000 expense; its cost is normally allocated over its useful life under the applicable accounting rules.

2.4 Double-entry bookkeeping

Double-entry bookkeeping records every transaction using entries whose total debits equal total credits. A *debit* is an entry on the left side of an account and a *credit* is an entry on its right side. These terms do not mean “debt” and “income”. Debits normally increase assets and expenses; credits normally increase liabilities, equity and revenue. Entries in the opposite direction decrease these accounts.

Equal debits and credits do not mean that every transaction changes both sides of the accounting equation. A purchase paid in cash exchanges one asset for another. When a supplier payable is settled, the payable is debited and cash is credited: both account balances decrease while the entry remains balanced.

2.5 Profit and cash collection

Suppose a product with a recorded cost of EUR 500 is sold for EUR 1,000 on 60-day credit and there are no other relevant expenses. The company records EUR 1,000 of revenue, EUR 500 of cost of sales and EUR 500 of profit. It also records a EUR 1,000 receivable from the customer. Cash collected from this sale remains zero until the customer pays. The later payment converts the receivable into cash; it does not create a second EUR 1,000 of revenue.

2.6 Bookkeeping, accounting and account families

Bookkeeping records and organises transactions. A journal, also called a daybook, records entries in chronological order. A general ledger groups entries by account and shows their movements and balances. A chart of accounts lists the accounts used by the business. Accounting also involves classifying, adjusting, summarising, reporting and interpreting the information. Bookkeeping therefore supports accounting; their difference is broader than simply “what happened” versus “why it happened”.

The five basic account families are:

Family	Meaning and examples	Normally increases with
Assets	Controlled resources, such as cash, equipment and receivables	Debit
Liabilities	Obligations, such as loans, supplier payables and taxes payable	Credit
Equity	Owners’ residual interest, including capital and retained earnings	Credit
Revenue	Income from ordinary activities, such as sales and services	Credit
Expenses	Recognised costs, such as wages, rent and cost of sales	Debit

Revenue and expenses explain the profit or loss over a period; their net effect contributes to equity. Assets, liabilities and equity appear in the balance sheet at a date, whereas revenue and expenses appear in the income statement for a period. The left and right sides of an individual account are not instructions about how to lay out a financial statement.

Chapter 3

Financial analysis

3.1 Purpose and main areas

Financial analysis examines a company's position and performance using financial statements and other information. It helps managers, investors and creditors assess liquidity, solvency, profitability and efficiency. Analysts use ratios, changes over time and comparisons with similar businesses. A single ratio provides a clue, not a complete judgment.

Liquidity concerns the ability to meet obligations due in the short term. Solvency concerns the ability to sustain operations and meet financial commitments over the longer term. Profitability relates income to sales or the resources used. Efficiency describes how well resources generate sales; asset turnover is one measure of efficiency, not of profit by itself.

3.2 Liquidity: working capital and ratios

Cash is immediately available for payment, but other current assets may not be converted into cash quickly or at their recorded value. A profitable company can therefore experience a cash shortage.

Current assets include cash and items expected to be sold, consumed or realised in the normal operating cycle, or generally within twelve months. Current liabilities are obligations expected to be settled in that cycle or due in the short term. The twelve-month rule is a useful simplification, not the only classification criterion. Equity is not a current liability and is not subtracted when calculating working capital.

Net working capital is an amount expressed in currency:

$$\text{Working capital} = \text{Current assets} - \text{Current liabilities.}$$

A positive value means current assets exceed current liabilities at the balance-sheet date. It does not ensure that cash will be available when each bill becomes due: receivables may be collected late and inventory may sell slowly.

The current ratio compares the same categories as working capital but expresses their relationship as a quotient:

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}.$$

The quick ratio focuses on assets that can more readily be used to pay short-term obligations:

$$\text{Quick ratio} = \frac{\text{Cash} + \text{Short-term investments} + \text{Receivables}}{\text{Current liabilities}}.$$

It excludes inventory and other items, such as prepaid expenses, that cannot readily pay bills. An alternative simplified numerator is current assets minus inventory and prepayments, when those categories explain the difference. A ratio above 1 says that the assets included in its numerator exceed current liabilities on that date; it does not guarantee timely payment. The quality of receivables, payment dates, cash flows and industry norms also matter. Neither ratio is defined in the usual way when current liabilities are zero.

3.3 Solvency and indebtedness

Solvency depends on obligations, the value and composition of assets, and the capacity to generate future cash. Merely having total assets above long-term debt is not a sufficient assessment. Borrowing can finance growth but creates payment commitments that increase risk if income or cash flows fall.

The term “financial leverage ratio” is ambiguous: it can refer to different measures. If the numerator includes *all liabilities*, the first ratio should be labelled liabilities-to-equity:

$$\text{Liabilities-to-equity} = \frac{\text{Total liabilities}}{\text{Total equity}}.$$

A value of 0.8 means that the company has 80 cents of liabilities for each euro of book equity. Some analysts use only interest-bearing borrowing in the numerator and call the result debt-to-equity; this is a different definition. The equity multiplier, assets divided by equity, is another leverage measure. A debt-to-equity figure is difficult to interpret in the usual way when equity is zero or negative.

If “debt” is defined as all liabilities, the corresponding ratio to assets is

$$\text{Liabilities-to-assets} = \frac{\text{Total liabilities}}{\text{Total assets}}.$$

This indicates the share of book assets financed through obligations to others. If a course uses *debt ratio* for interest-bearing borrowing divided by assets, that narrower definition should be stated and used consistently. A higher ratio can indicate greater financial risk, but its meaning depends on the business model, cash-flow stability and comparison group.

3.4 Operating efficiency: asset turnover

Total asset turnover measures sales generated per unit of assets during a period:

$$\begin{aligned}\text{Total asset turnover} &= \frac{\text{Net sales}}{\text{Average total assets}}, \\ \text{Average total assets} &= \frac{\text{Opening assets} + \text{Closing assets}}{2}.\end{aligned}$$

A high turnover can reflect effective use of assets, but it does not show the profit margin earned on those sales. Because net sales cover a period and the balance sheet measures assets at points in time, the average of opening and closing assets is commonly used.

3.5 Earnings: EBIT and EBT

EBIT means *earnings before interest and taxes*. It is an amount, not a ratio, and is commonly used to examine performance before financing cost and income tax. EBT means *earnings before taxes*: it reflects interest expense and other relevant items before income tax. In a simplified income statement, EBIT minus net interest expense gives EBT, and EBT minus income tax gives net income. Actual statements can contain other non-operating items, so this sequence needs to be checked against the statement being analysed.

3.6 Returns: ROI, ROA and ROE

For a particular project, one simple definition of return on investment is

$$\text{ROI} = \frac{\text{Net gain from the investment}}{\text{Cost of the investment}} \times 100\%.$$

Net gain means proceeds or benefits minus the investment cost and any other costs included in the chosen definition. If proceeds are 120 and the investment costs 100, net gain is 20 and ROI is 20%. Managerial accounting may instead use “ROI” for a segment’s operating income divided by its average invested assets. Its numerator, denominator and period must therefore be specified. A simple ROI also ignores when cash flows occur.

Using net income as the measure of profit, return on assets and return on equity can be written as

$$\text{ROA} = \frac{\text{Net income}}{\text{Average total assets}} \times 100\%,$$
$$\text{ROE} = \frac{\text{Net income}}{\text{Average equity}} \times 100\%.$$

The averages ordinarily use opening and closing balances. ROA measures net income relative to the assets employed; ROE measures net income relative to owners' equity. Some courses use an operating-profit version of ROA: the two versions should not be mixed in a comparison. Debt can amplify ROE and also amplify losses, so a high ROE alone does not establish that a business is stronger. ROE becomes hard to interpret when equity is very small or negative.

3.7 Interpreting comparisons

EBIT and EBT are absolute earnings amounts, while ROI, ROA and ROE are ratios. Ratios can help compare businesses of different sizes, and EBIT can also be compared across businesses when its definition and the underlying accounting are aligned. All these measures can be examined over time. Useful comparisons require compatible definitions, the same period and, generally, businesses in similar industries. No metric is automatically more useful than the others in every situation.

3.8 References for the definitions

The accounting definitions and ratio conventions in these notes were checked against the IFRS Conceptual Framework, IAS 7, and the OpenStax sections on liquidity, solvency, asset turnover and profitability.

Chapter 4

Worked example: from transactions to financial ratios

This is a fictional company. All amounts in this chapter are in **thousands of euros**: the number 20 means EUR 20,000. We examine one year, from 1 January to 31 December. There are no short-term investments or prepaid expenses. For simplicity, interest and income tax are paid during the year, and interest paid is included in operating cash flows.

4.1 Opening balance sheet

At the start of the year, the company has the following balances:

Assets	Amount	Liabilities and equity	Amount
Cash	20	Supplier payables	10
Receivables from customers	10	Short-term bank loan	5
Inventory	15	Long-term bank loan	35
Equipment, net of depreciation	55	Share capital	40
		Retained earnings	10
Total assets	100	Total liabilities and equity	100

The accounting equation holds: assets of 100 equal liabilities of 50 plus equity of 50.

4.2 Transactions and journal entries during the year

All sales in this simplified example are initially made on credit; customer collections include both opening receivables and this year's sales. Inventory purchases are also made on credit; supplier payments include the opening payable. Each transaction has equal debits and credits:

Transaction	Debit	Credit
Sell goods on credit for 120	Customer receivables 120	Sales revenue 120
Collect 110 from customers	Cash 110	Customer receivables 110
Buy inventory on credit for 75	Inventory 75	Supplier payables 75
Pay suppliers 73	Supplier payables 73	Cash 73
Recognise cost of goods sold of 72	Cost of goods sold 72	Inventory 72
Pay wages, rent and other operating costs of 25	Operating expenses 25	Cash 25
Recognise depreciation of 5	Depreciation expense 5	Accumulated depreciation 5
Pay interest of 3	Interest expense 3	Cash 3
Pay income tax of 4	Income tax expense 4	Cash 4
Buy new equipment for 10 in cash	Equipment 10	Cash 10
Receive a new long-term loan of 8	Cash 8	Long-term loan 8
Receive a capital contribution of 5	Cash 5	Share capital 5
Pay a dividend of 2	Retained earnings 2	Cash 2

For instance, a sale on credit increases revenue and receivables, but does not increase cash on the sale date. Depreciation is an expense that reduces the net carrying amount of equipment without a cash payment in this year. The new loan and owner's contribution increase cash without creating revenue.

4.3 Income statement for the year

Only revenue and expenses recognised for this year determine profit:

Item	Amount
Sales revenue	120
Cost of goods sold	(72)
Gross profit	48
Cash operating expenses	(25)
Depreciation expense	(5)
EBIT, earnings before interest and taxes	18
Interest expense	(3)
EBT, earnings before taxes	15
Income tax expense	(4)
Net income	11

Thus EBIT is $120 - 72 - 25 - 5 = 18$, EBT is $18 - 3 = 15$, and net income is $15 - 4 = 11$. Neither the loan of 8 nor the capital contribution of 5 is revenue. Buying equipment for 10 is not an immediate operating expense of 10 in this example; depreciation of 5 is the expense recognised this year.

4.4 Closing balance sheet

We first update each balance using opening balance plus additions minus reductions:

$$\begin{aligned}
 \text{Receivables} &= 10 + 120 - 110 = 20, \\
 \text{Inventory} &= 15 + 75 - 72 = 18, \\
 \text{Equipment, net} &= 55 + 10 - 5 = 60, \\
 \text{Supplier payables} &= 10 + 75 - 73 = 12, \\
 \text{Long-term loan} &= 35 + 8 = 43.
 \end{aligned}$$

Cash is calculated from the actual payments and receipts, not from net income:

$$\text{Closing cash} = 20 + 110 - 73 - 25 - 3 - 4 - 10 + 8 + 5 - 2 = 26.$$

Share capital becomes $40 + 5 = 45$. Retained earnings become $10 + 11 - 2 = 19$: opening retained earnings plus this year's net income minus the dividend. Consequently, the closing balance sheet is:

Assets	Amount	Liabilities and equity	Amount
Cash	26	Supplier payables	12
Customer receivables	20	Short-term bank loan	5
Inventory	18	Long-term bank loan	43
Equipment, net	60	Share capital	45
		Retained earnings	19
Total assets	124	Total liabilities and equity	124

The equation still holds: $124 = 60 + 64$. The dividend reduced cash and retained earnings; it was not an expense in the income statement.

4.5 Cash flow statement for the year

The cash flow statement reconciles opening cash of 20 to closing cash of 26:

Cash flow	Amount
Collections from customers	110
Payments to suppliers	(73)
Operating expenses paid	(25)
Interest paid	(3)
Income tax paid	(4)
Net cash from operating activities	5
New equipment purchased (investing)	(10)
New long-term loan received (financing)	8
Owner contribution (financing)	5
Dividend paid (financing)	(2)
Net change in cash	6
Opening cash	20
Closing cash	26

Operating cash is $110 - 73 - 25 - 3 - 4 = 5$; investing cash is -10 ; financing cash is $8 + 5 - 2 = 11$. Their sum is $5 - 10 + 11 = 6$.

The difference between net income of 11 and operating cash of 5 is also visible through the balance-sheet movements:

$$\underbrace{11}_{\text{Net income}} + \underbrace{5}_{\text{Depreciation}} - \underbrace{10}_{\text{Increase in receivables}} - \underbrace{3}_{\text{Increase in inventory}} + \underbrace{2}_{\text{Increase in payables}} = 5.$$

The company reported a profit of 11 but collected only 5 in net operating cash, partly because receivables and inventory increased. For this illustration, interest paid was grouped with operating payments; classification rules and policies should be checked when analysing a real cash flow statement.

4.6 Calculating the financial ratios

The closing *current assets* are cash 26 plus receivables 20 plus inventory 18, or 64. The closing *current liabilities* are supplier payables 12 plus the short-term bank loan 5, or 17. The quick assets are cash and receivables, or 46. Using the closing balance sheet:

Working capital = $64 - 17 = 47$ thousand euros,

$$\text{Current ratio} = \frac{64}{17} = 3.76,$$

$$\text{Quick ratio} = \frac{26 + 20}{17} = 2.71,$$

$$\text{Liabilities-to-equity} = \frac{12 + 5 + 43}{45 + 19} = \frac{60}{64} = 0.94,$$

$$\text{Liabilities-to-assets} = \frac{60}{124} = 48.39\%.$$

The two indebtedness ratios above count *all liabilities*. If only interest-bearing bank borrowing is included, the debt-to-equity ratio instead equals $(5 + 43)/64 = 0.75$. These are different measures and should not be given the same label without defining “debt”.

To compare income or sales earned *during the year* with assets and equity, use their average opening and closing balances:

$$\text{Average assets} = \frac{100 + 124}{2} = 112,$$

$$\text{Average equity} = \frac{50 + 64}{2} = 57,$$

$$\text{Asset turnover} = \frac{120}{112} = 1.07,$$

$$\text{ROA} = \frac{11}{112} \times 100 = 9.82\%,$$

$$\text{ROE} = \frac{11}{57} \times 100 = 19.30\%.$$

Asset turnover of 1.07 means the company generated about EUR 1.07 of sales per EUR 1 of average assets. ROA relates net income to average assets, while ROE relates the same net income to average equity. ROE is higher partly because assets are financed by liabilities as well as equity; it is not enough by itself to judge whether the company is financially safe. As an additional measure, net profit margin is $11/120 \times 100 = 9.17\%$.

4.7 ROI for a particular investment

ROI cannot be determined for the equipment purchase from the closing balance sheet alone: the example does not separately identify the benefits and costs attributable to that equipment. In a *separate hypothetical project*, suppose an investment costs 10 and produces total net proceeds of 12. Its net gain is $12 - 10 = 2$, so the simple project ROI is

$$\text{ROI} = \frac{12 - 10}{10} \times 100 = 20\%.$$

This project example is independent of the company's statements above. A meaningful investment decision would also consider when cash arrives and the risk of receiving it.